

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Right triangles and trigonometry

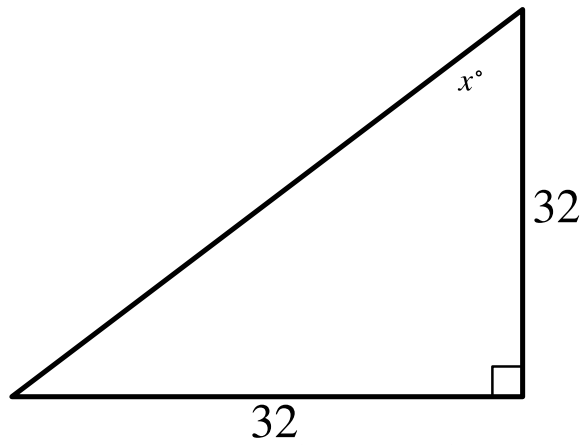
02

4 questions · ~6 min

Q1

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 32
 - 32
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

$\overline{\textcircled{9}}$ $\overline{\textcircled{9}}$ $\overline{\textcircled{9}}$ $\overline{\textcircled{9}}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

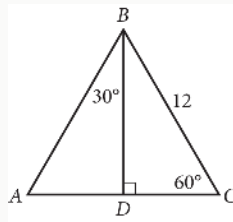
Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin F = \frac{308}{317}$, what is the value of $\sin(J)$?

A. $\frac{75}{317}$

B. $\frac{308}{317}$

C. $\frac{317}{308}$

D. $\frac{317}{75}$



In $\triangle ABC$ above, what is the length of $\overline{AD}$?

A. 4

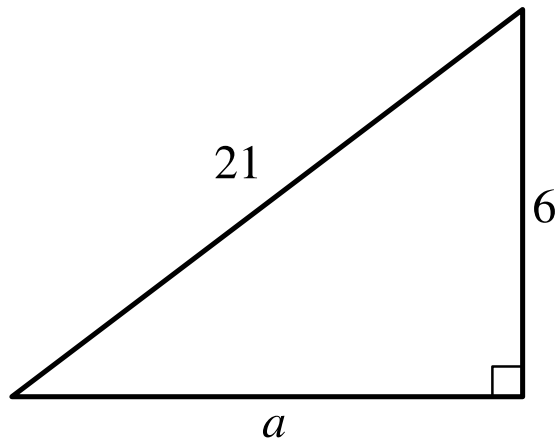
B. 6

C.

$$6\sqrt{2}$$

D.

$$6\sqrt{3}$$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 21.
- The length of one side adjacent to the right angle is 6.
- The length of the other side adjacent to the right angle is a .
- A note indicates the figure is not drawn to scale.

For the triangle shown, which expression represents the value of a ?

- A. $\sqrt{21^2 - 6^2}$
- B. $21^2 - 6^2$
- C. $\sqrt{21 - 6}$
- D. $21 - 6$